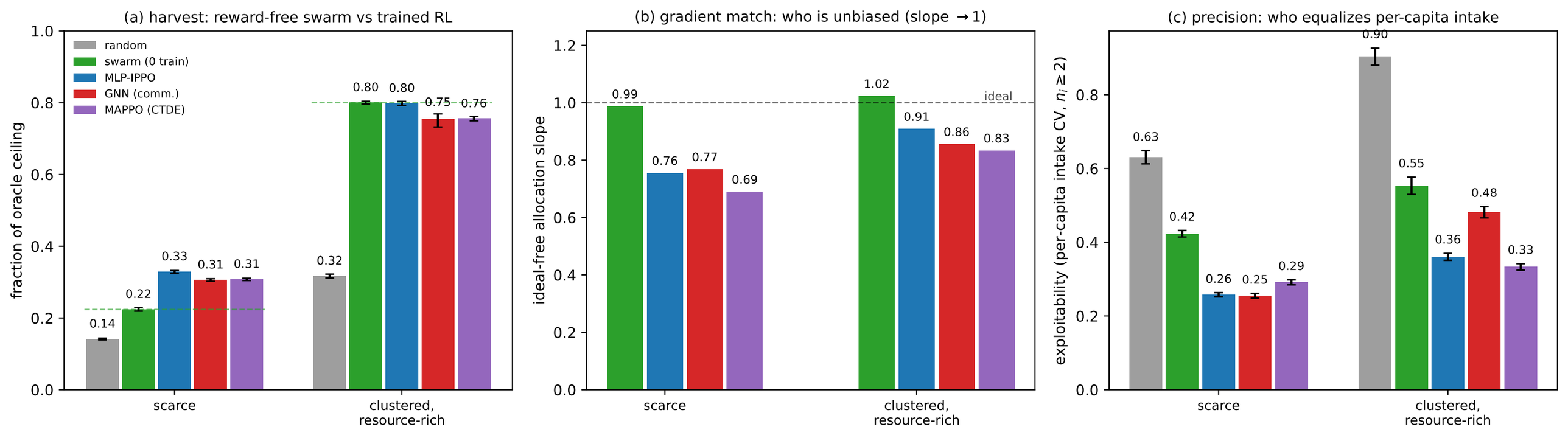
(a) harvest: reward-free swarm vs trained RL
fraction of oracle ceiling
random
swarm (0 train)
MLP-IPPO
GNN (comm.)
MAPPO (CTDE)
0.14
0.22
0.33
0.31
0.31
0.32
0.80
0.80
0.75
0.76
scarce
clustered, resource-rich
(b) gradient match: who is unbiased (slope → 1)
ideal-free allocation slope
ideal
0.99
0.76
0.77
0.69
1.02
0.91
0.86
0.83
scarce
clustered, resource-rich
(c) precision: who equalizes per-capita intake
exploitability (per-capita intake CV, $n_i \geq 2$)
0.63
0.42
0.26
0.25
0.29
0.90
0.55
0.36
0.48
0.33
scarce
clustered, resource-rich